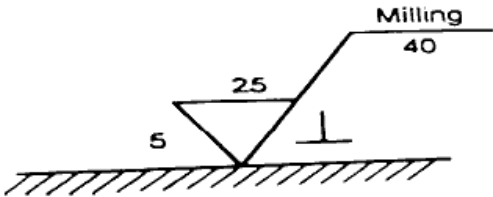


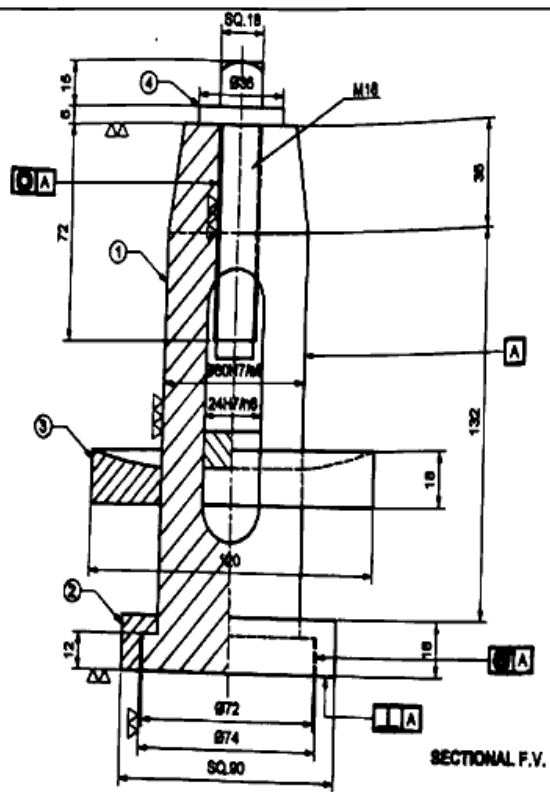
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Summer Examination – 2025****Course: B. Tech****Branch : Mechanical Engineering****Semester : III****Subject Code & Name: BTMEC304_Y19 , Machine Drawing and CAD****Max Marks: 60****Date: 24-07-2025****Duration: 04 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	For the object which is symmetrical about the central axis; from which method of section it can be drawn	(1/1)	1				
	<table> <tr> <td>a. Full section</td><td>b. Half section</td><td>c. Revolved section</td><td>d. Removed section</td></tr> </table>	a. Full section	b. Half section	c. Revolved section	d. Removed section		
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2	When the section is to be drawn for a small area of the object, a section is used.	(1/1)	1				
	<table> <tr> <td>a. Full section</td><td>b. Half section</td><td>c. Revolved section</td><td>d. . Removed section</td></tr> </table>	a. Full section	b. Half section	c. Revolved section	d. . Removed section		
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3	which of the following coupling is used to connect two parallel shafts when they are at a small distance apart?	(1/3)	1				
	<table> <tr> <td>a. Hooke's coupling</td><td>b. Rigid coupling</td><td>c. Muff coupling</td><td>d. Oldham's coupling</td></tr> </table>	a. Hooke's coupling	b. Rigid coupling	c. Muff coupling	d. Oldham's coupling		
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4	What are the two types of rigid flange couplings in machine design	(1/3)	1				
	<table> <tr> <td>a. Stable and unstable</td><td>b. Flexible and non-flexible</td><td>c. Protected and unprotected</td><td>d. Deformable and non-deformable</td></tr> </table>	a. Stable and unstable	b. Flexible and non-flexible	c. Protected and unprotected	d. Deformable and non-deformable		
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5	When The line of intersection formed by two solids is curve then the solids are	(1/2)	1				
	<table> <tr> <td>a. Cone, Cylinder</td><td>b. Cube, Prism</td><td>c. Pyramid, Cube</td><td>d. Pyramid, Cuboid</td></tr> </table>	a. Cone, Cylinder	b. Cube, Prism	c. Pyramid, Cube	d. Pyramid, Cuboid		
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6	When a solid completely penetrates another solid, there will be two lines of intersection. These lines are called.....	(1/2)	1				
	<table> <tr> <td>a. line of interpenetration</td><td>b. concyclic curves of lines</td><td>c. hidden lines</td><td>d. inside line</td></tr> </table>	a. line of interpenetration	b. concyclic curves of lines	c. hidden lines	d. inside line		
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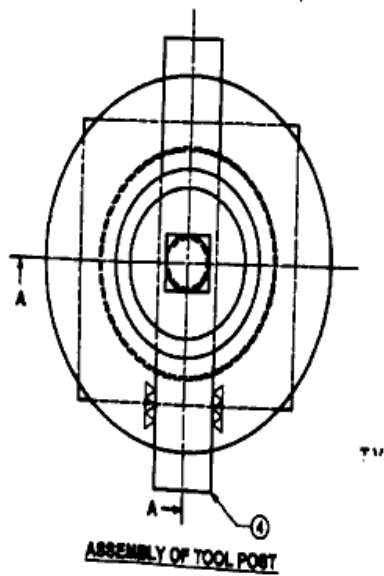
7	Drawing showing the position of each part with respect to each other is called.....				(1/4)	1
	a. assembly drawing	b. part drawing	c. machine drawing	d. installation drawing		
8	Detailed drawing of each part of a machine is called.....				(1/4)	1
	a. assembly drawing	b. part drawing	c. patent drawing	d. tabular drawing		
9	Which is an authorized document to produce a part in the workshop.				(1/5)	1
	a. Assembly drawing	b. Exploded drawing	c. Part drawing	d. Production drawing		
10	The relation between mating parts is called				(1/5)	1
	a. Connection	b. Fits	c. Joints	d. Link		
11	In AutoCAD 2D Modelling, which axis is not accessible for drafting				(1/6)	1
	a. X	b. Y	c. Z	d. WCS		
12	Which of the following is the shortcut for zoom command.				(1/6)	1
	a. Z Command	b. Rotate mouse wheel	c. ZOOM Command	d. All of the above		
Q. 2	Solve the following.					12
A)	Illustrate the following with an example a) Full section b) Removed section				(1/1)	6
B)	Draw the symbols of the following a) Flat double V-butt weld b) Concave fillet weld				(2/1)	6
Q.3	Solve the following.					12
A)	Show protected type flange coupling with a neat diagram.				(2/3)	6
B)	Represent single riveted joint.				(2/3)	6

Q. 4	Solve Any Two of the following.		12
A)	A vertical cylinder of 75 mm diameter is penetrated by another cylinder of 50 mm diameter, the axis of which is parallel to both the H.P. and the V.P. The two axes are 9 mm apart. Draw the projections of two cylinders showing curves of intersection. Assume suitable axis lengths for both the cylinders.	(3/2)	6
B)	A vertical square prism, base 50 mm side and height 90 mm is completely penetrated by a horizontal square prism, base 35 mm side and axis length 90 mm so that their axes are 6 mm apart. The axis of the horizontal prism is parallel to the V.P., while the faces of both prisms are equally inclined to the V.P. Draw the projections of the prisms showing lines of intersection.	(3/2)	6
C)	A cone with base diameter 70 mm and axis height 65 mm is kept on the H.P. on its base. It is penetrated by a horizontal cylinder of diameter 35 mm with its axis parallel to V.P. and intersecting the axis of the cone at a distance of 20 mm above the base of the cone. Draw the projections of solids showing curves of intersection.	(3/2)	6
Q.5	Solve Any Two of the following.		12
A)	The hole has size ϕ 20.018 mm and ϕ 20.000 mm and shaft size ϕ 20.023 mm and ϕ 20.012 mm. Determine the types of fit between them.	(3/5)	6
B)	State the meaning of machining symbol as shown in figure	(2/5)	6
 The diagram shows a machining symbol on a hatched surface. It consists of a triangle with a 25-degree angle, a horizontal line segment of 5 mm, and a vertical line segment of 40 mm. The word 'Milling' is written above the vertical segment.			
C)	Write advantages of CAD and list out the commands in Draw.	(2/6)	6
Q. 6	Solve Any One of the following.		12
A)	Figure shows assembly of lathe tool post. Draw the detailed drawing of the following: a) Post – Sectional Front View and Top View b) Ring - Sectional Front View and Top View c) Block - Sectional Front View and Top View	(3/4)	12



PART LIST

PART NO.	PART NAME	MATL	QTY.
1	POST	C.I.	1
2	BLOCK	C.I.	1
3	RING	C.I.	1
4	SCREW	M.S.	1
5	WEDGE	M.S.	1
6	TOOL	H.S.S.	1

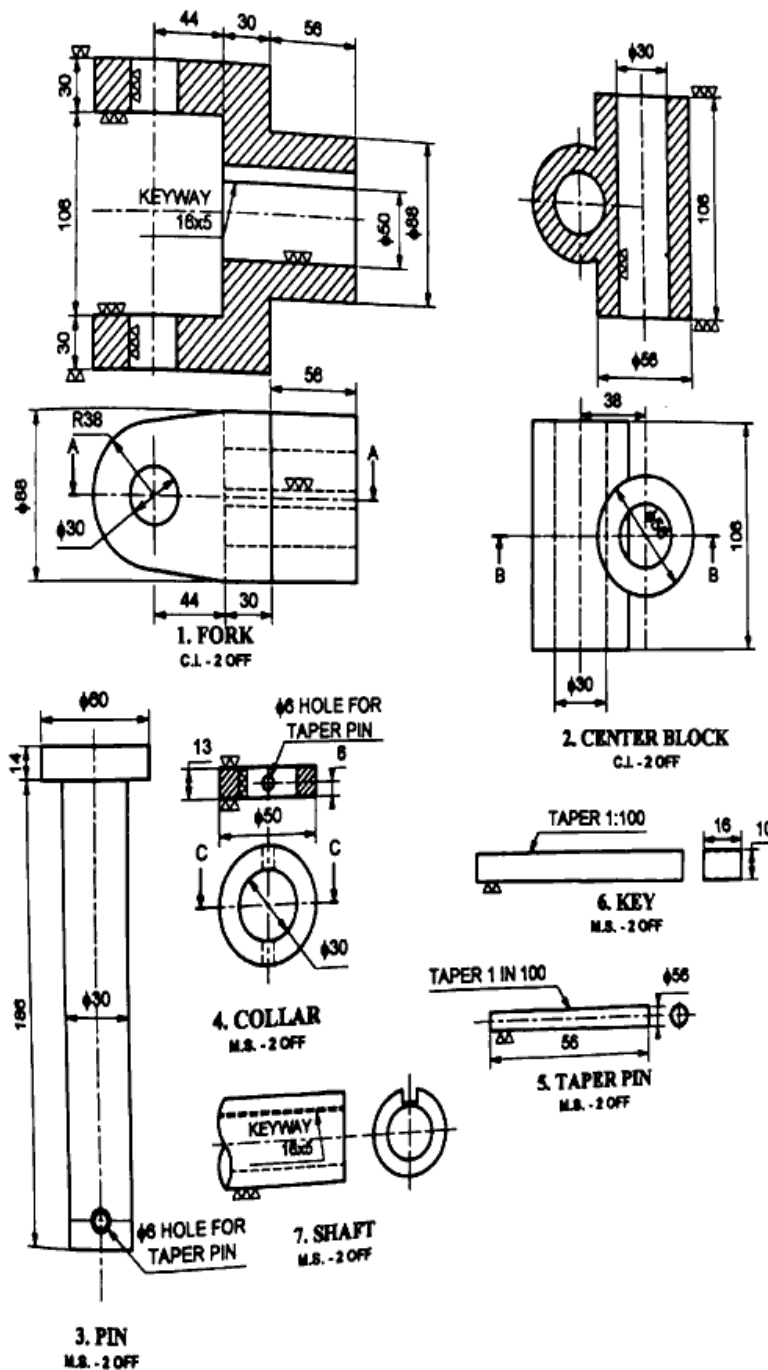


ASSEMBLY OF TOOL POST

B) Figure shows details of universal coupling. Draw sectional front view and top view of the assembly and also prepare bill of material.

(3/4)

12



*** End ***